Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

We compare gravity, drag, constant net force and a gravity–drag–Magnus model. M3 estimates constant acceleration from early positions; M4 fits a constant drag coefficient and a lift coefficient proportional to the spin parameter.

Three recorded penalties provide 39 tracking frames. After conversion to penalty-mark coordinates, three selected positions per event enter 12 bounded inversions, compared with later samples. Two placement shots and one power shot are analyzed, comparing speed, aerodynamic forces and prediction error. A separate curved-shot calculation examines rotational damping. Section 5 uses the competition positions for individual perturbations, position-constrained crossings and 100,000 Monte Carlo trajectories.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

A kicked soccer ball can curve in flight: gravity pulls it downward, drag slows it, and rotation produces lateral or vertical force. Previous studies combine aerodynamic forces with motion equations and measured trajectories[5], and examine how drag, spin and wind affect three-dimensional flight[10, 8].

The ball's surface also affects its aerodynamic forces. Wind-tunnel measurements relate flow conditions to those forces[11], while simulations compare a seamed football with a smooth sphere[9]. These studies support a spherical approximation followed by an examination of force coefficients.

Measured flights provide a basis for comparison. Ball markings and cameras can track position and spin[3], while trajectories can be used to estimate drag and lift coefficients[15, 4].

The problem supplies an initial position and two later positions, but not initial velocity or spin. We find states compatible with those observations and calculate their goal-plane crossings. Recorded match flights provide a comparison; Appendix A gives the source and processing.

1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions relate to the unknown motion reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 examines these sensitivities and joint perturbations.

We estimate initial states from three real penalties and compare placement and power shots with later recorded positions. Figure 1 summarizes this physical and computational sequence.

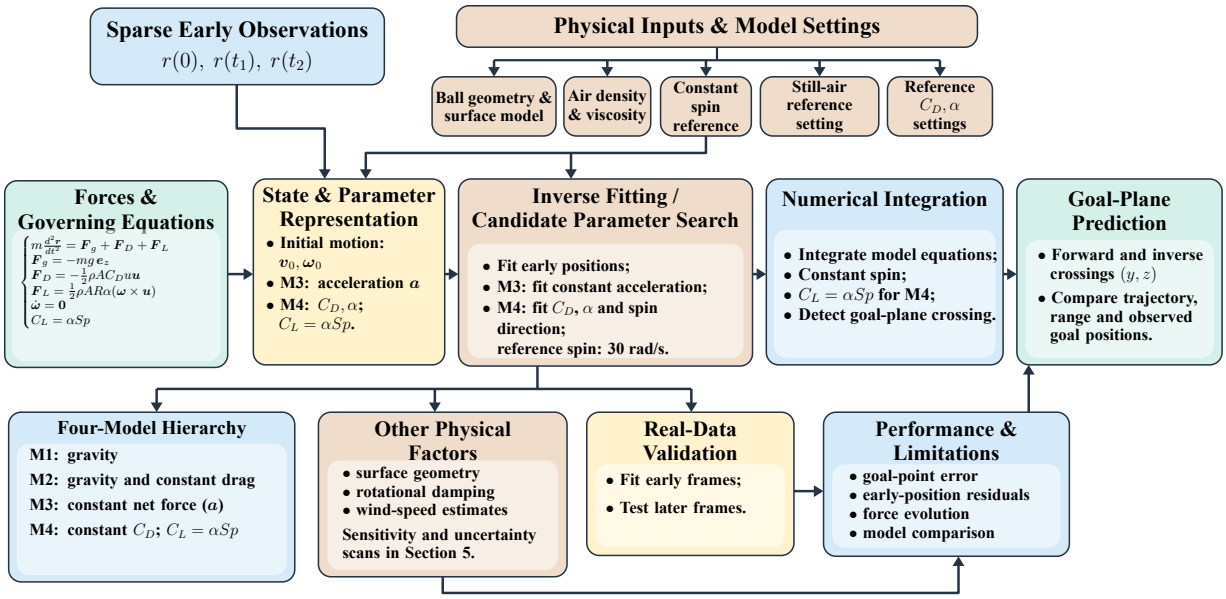


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

- **Contact and model start.** Real inversions define $t = 0$ at the selected starting tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [8, 5, 10]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [8, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.
- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.

- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 uses the competition-position example to examine parameter effects on crossing distributions.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The ball and air parameters use published reference values [8, 11]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with x towards the goal, y lateral and z upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m. Section 4.1 converts the measured pitch-centred coordinates.

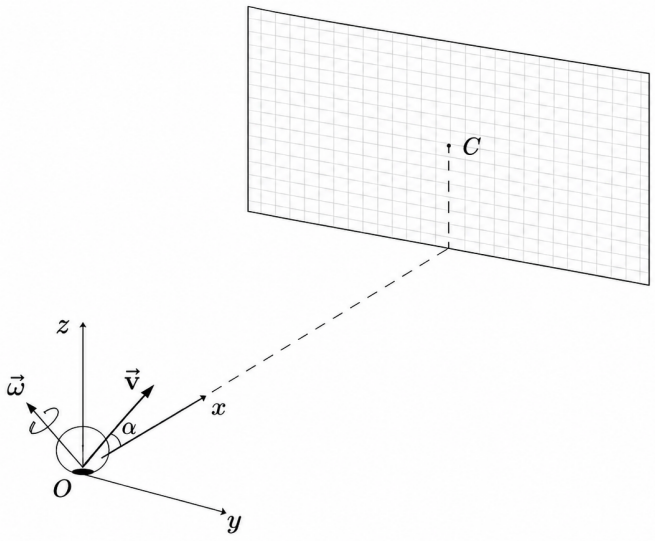


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

A clean opening crossing requires $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about 1.56% of the ball's weight and changes the downward acceleration from g to $g(1 - \rho V_b/m)$.

The 1.56% buoyancy scale is a small vertical correction; all four main comparisons use gravity, drag and Magnus lift as defined above.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.

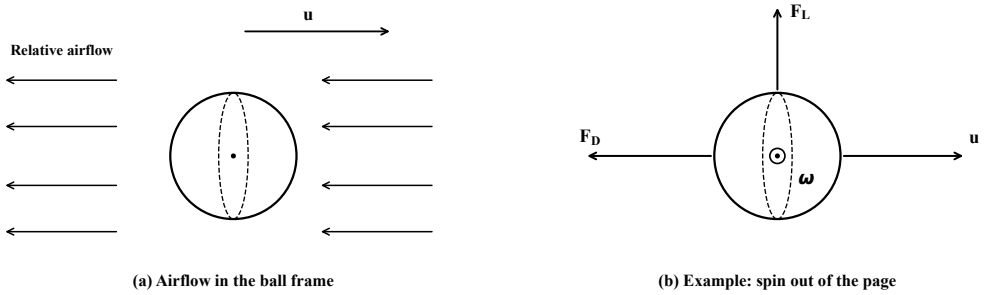


Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.2 Aerodynamic Drag

With relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$, $u = \|\boldsymbol{u}\|$ and $A = \pi R^2$, drag is [15, 12]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

The drag coefficient C_D is constant during each flight.

3.2.3 Magnus Lift

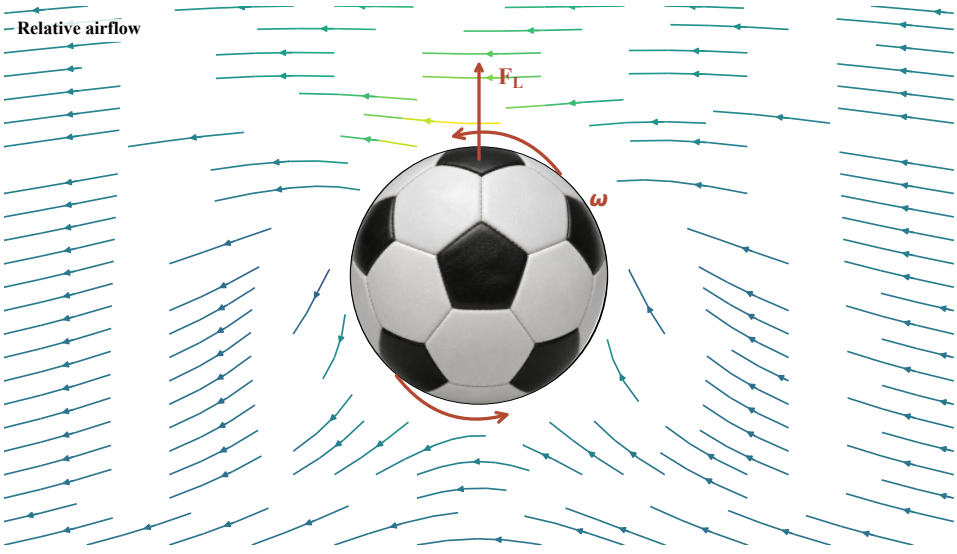


Figure 4: Idealized flow around a spinning football.

Let $Sp = R\|\boldsymbol{\omega}\|/u$ and $C_L = \alpha Sp$. Resolving the transverse component yields

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The force vanishes at zero spin or zero relative velocity; reversing cross-product order requires reversing the sign. The forces on the football are summarized in the following diagram.

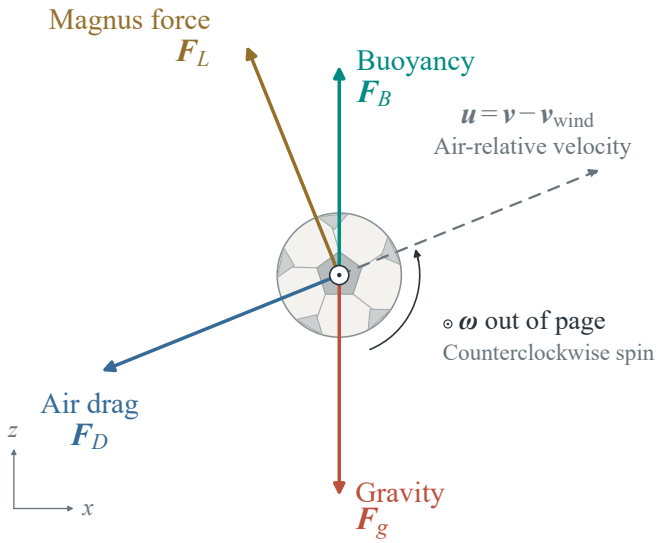


Figure 5: Forces on the football: gravity, buoyancy, drag and Magnus force, with arrows showing their directions for the illustrated state.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b \boldsymbol{\omega}. \tag{5}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

We infer initial motion from three recorded penalties and compare the flight and prediction of placement and power shots.

Table 2: Models and inverse quantities

Model	Assumption	Inverse quantities
M1	gravity	velocity (3)
M2	gravity and drag	velocity, drag coefficient (4)
M3	constant net force	velocity, acceleration (6)
M4	gravity, drag, Magnus force	velocity, spin direction, C_D, α (7)

M3 represents the combined net force. M4 uses constant C_D and α during each flight with $C_L = \alpha Sp$. Forward reference values are $C_D^* = 0.2415$ and $\alpha^* = 1.5251$ at a published reference condition [12].

4.1 Inverse Prediction from Three Real Penalties

4.1.1 Observations and prediction

We test the models on three recorded penalties: two placement shots and one power shot. The first selected position defines the time origin, and the positions at 0.04 and 0.08 s complete the input for predicting the flight. See Appendix A for the data source and frame selection.

The horizontal origin is the penalty mark, x points toward the goal, y is lateral and z is ball-centre height; the goal plane is $x = 11$ m. Later positions are used for comparison, and the observed goal crossing is taken from a measurement or interpolation between adjacent positions.

4.1.2 Bounded inversion and selection

M3 determines constant acceleration from three positions:

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{8}$$

M4 minimizes the position error over velocity, spin direction, C_D and α. Six deterministic initial guesses are used within the ranges in Table 9 in Appendix A; the minimum-residual solution is retained. Only the product $\alpha\boldsymbol{\omega}$ enters Magnus force, so spin magnitude and α are not separately identified by positions. A reference spin magnitude of 30 rad/s is assumed. Seven fitted quantities and six later position coordinates leave multiple possible solutions.

4.2 Case Analysis and Validation

Figure 6 illustrates the two shot styles: the placement-shot ball is close to the turf, while the power-shot ball is higher in the air.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

The recorded penalties comprise two placement shots and one power shot. Figure 7 compares their spatial trajectories. Open circles indicate fitting inputs, black lines and points denote tracking, colored lines show the models, and crosses mark measured or interpolated goal crossings. Predictions stop at first ground contact.

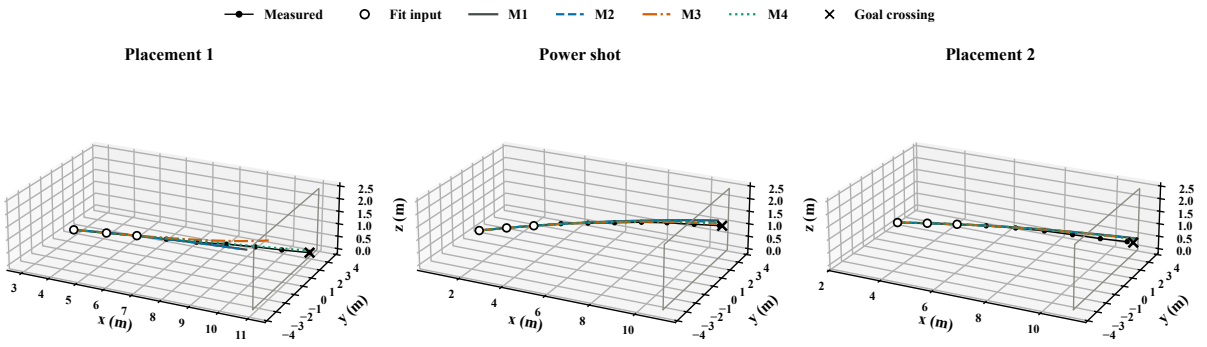


Figure 7: Three recorded penalty trajectories and model predictions, from left to right: placement 1, power shot, placement 2. Each inversion uses positions at 0, 0.04 and 0.08 s after the selected start.

4.2.1 Case 1: Placement Shots

As shown in Figure 6a, the placement-shot illustration shows the ball close to the turf. Figure 8 compares the two recorded placement trajectories with the four model predictions.

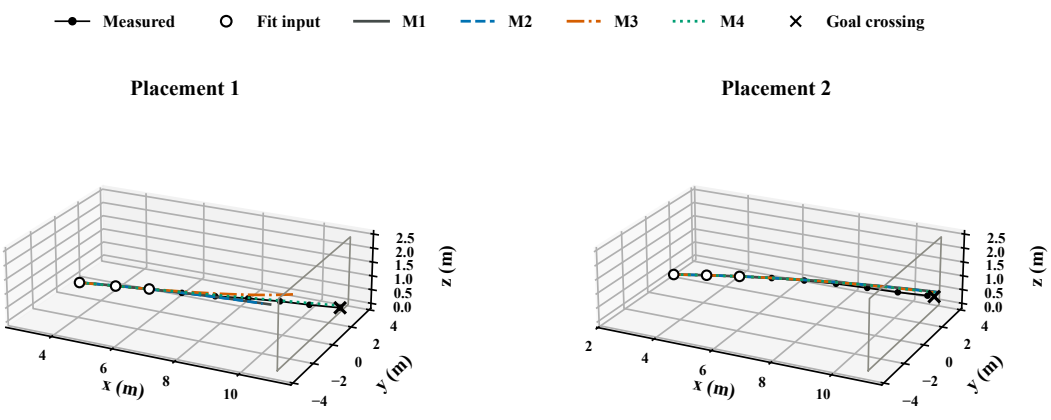


Figure 8: Measured and predicted trajectories for the two placement cases.

The two placement shots both travel close to the ground. M4 estimates their initial speeds at the selected frames as 29.20 and 28.34 m/s, respectively. For placement 1, M4 reaches the goal plane with a 0.360 m error; M1 and M2 hit the ground and M3 does not reach the goal within the calculation horizon. For placement 2, M1, M2, M3 and M4 all reach the goal, with errors of 0.475, 0.475, 0.155 and 0.146 m, respectively.

A low trajectory is sensitive to vertical forces. Gravity pulls downward, while the vertical Magnus component can raise or lower the path. M1 omits aerodynamic forces, and M2 adds drag without transverse lift. M3 fits a constant acceleration to the early points, but its continuation can deviate as acceleration evolves. M4 allows aerodynamic forces to change with velocity and spin direction; it reaches the goal for placement 1 and gives a smaller crossing error for placement 2. Early tracking and the fitted parameters both influence these differences.

4.2.2 Case 2: Power Shot

As shown in Figure 6b, the illustrative power shot places the ball well above the turf. M4 estimates the recorded power shot's speed at the selected start as 32.05 m/s, higher than for the two placement examples. Figure 9 shows its spatial path and x, y, z against time. The first three positions are fitted; later black points assess the continuation.

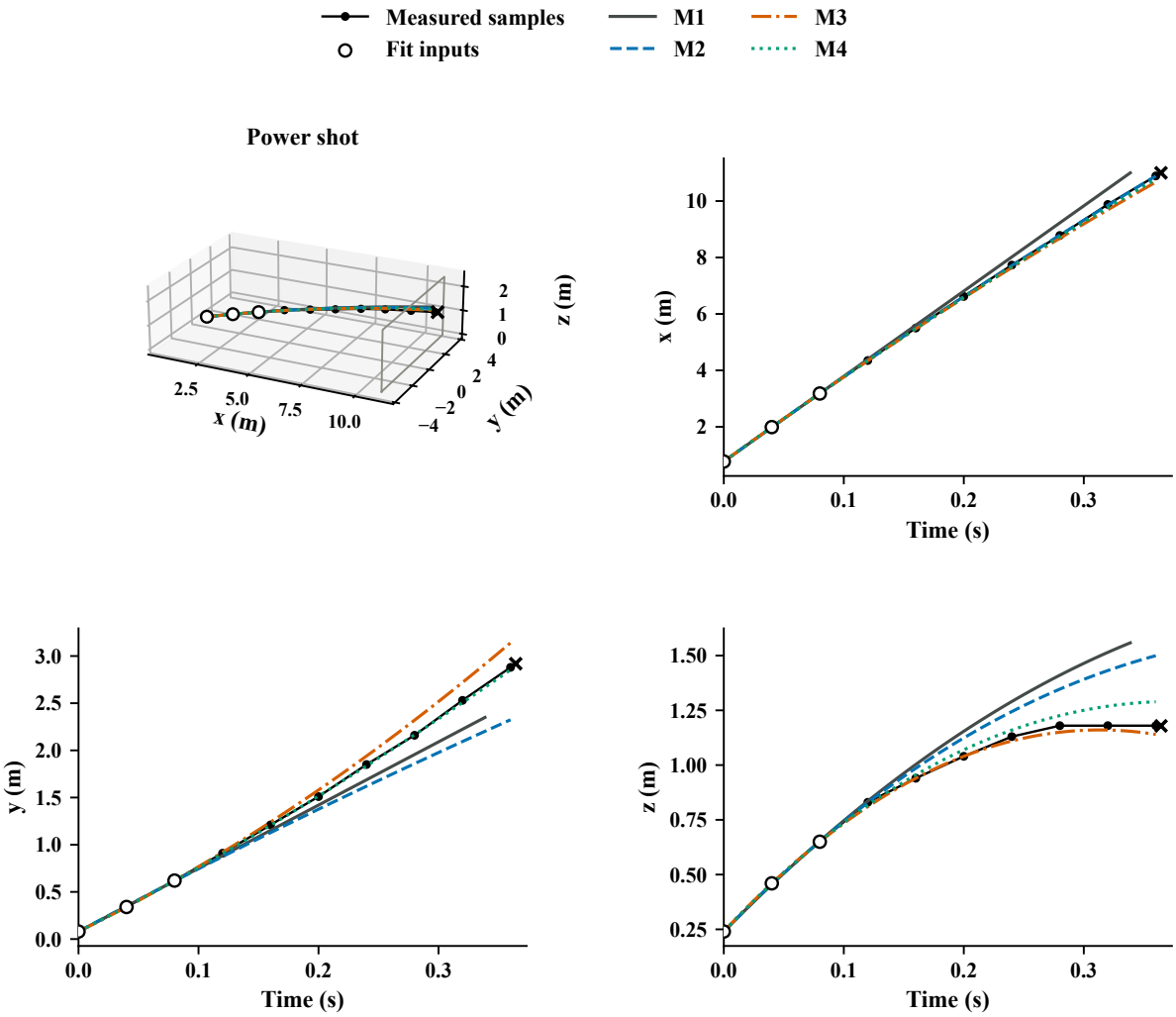


Figure 9: Power-shot trajectory and coordinates against time. Open circles are fitting inputs and black points are measured samples.

$x(t)$ shows forward deceleration; $y(t)$ and $z(t)$ show lateral and vertical curvature. M1, M2, M3 and M4 have goal-plane errors of 0.684, 0.657, 0.374 and 0.113 m, respectively. M1 uses gravity alone and M2 adds drag; their horizontal directions are mainly set by initial velocity. M3 continues the constant acceleration determined from the three inputs. M4 uses drag and Magnus force to describe evolving curvature and agrees more closely with the later points in this shot. Errors compare each model's own (y, z) crossing with the measured crossing.

Physical picture of speed and aerodynamic forces. Figure 10 compares M4 speed and force magnitudes. Drag scales with relative speed squared: the fitted initial drag is about 7.58 N for the power shot and 7.05 and 2.21 N for the placement shots, with coefficient values also affecting the comparison. As the ball slows, drag decreases. Magnus force depends on the cross product of spin and relative velocity: spin direction determines whether its effect is chiefly lateral or vertical, and its magnitude changes with the motion. The displayed speeds and forces are computed from the

M4 inversions, using a reference spin magnitude of 30 rad/s.

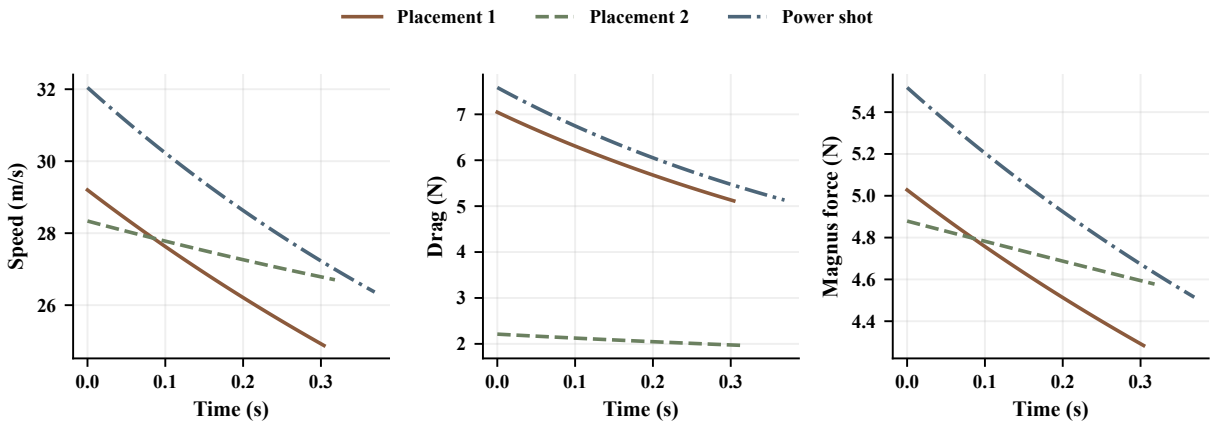


Figure 10: M4 speed, drag and Magnus force for the three recorded penalties. Curves end at first model-predicted ground contact or goal-plane crossing.

Table 3: Goal-plane errors (m); a dash denotes no valid goal-plane crossing.

Event	M1	M2	M3	M4
Placement 1	—	—	—	0.360
Power shot	0.684	0.657	0.374	0.113
Placement 2	0.475	0.475	0.155	0.146

5 Sensitivity Analysis

We retain the individual-parameter, early-position-constrained and Monte Carlo structure for the current M4 equations. The reference observations are $(0, 0, 0.11)$ m at 0 s, $(1.50, 0.45, 0.44)$ m at 0.05 s, and $(2.96, 0.86, 0.74)$ m at 0.10 s. Mass and radius are fixed at 0.43 kg and 0.11 m.

The fitted reference uses $\boldsymbol{v}_0 = (30.3784, 9.4074, 6.8974)$ m/s.

The reference spin is $\boldsymbol{\omega}_0 = (19.7434, 1.2060, -22.5553)$ rad/s, $C_D = 0.3623$ and $\alpha = 2.2877$. The crossing is $(2.6159, 1.9526)$ m at 0.398967 s. The 30 rad/s spin magnitude is a reference setting; the maximum early-coordinate residual is 0.000784 m and both coefficients reach the fitted upper bounds.

5.1 Effects of individual parameters

One parameter changes at a time: speed by $\pm 5\%$; spin magnitude, drag coefficient, lift parameter and air density by $\pm 10\%$; velocity and spin-axis angles by $\pm 2^\circ$. Each uses 41 values, totaling 369 flights. The table compares $\pm 1^\circ$ angles and $\pm 5\%$ scalar changes.

Table 4: Goal-plane displacement and arrival-time change under one-factor perturbations.

Parameter	Perturbation	max D_g (cm)	max $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag multiplier	$\pm 5\%$	0.620	2.274
Lift multiplier	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

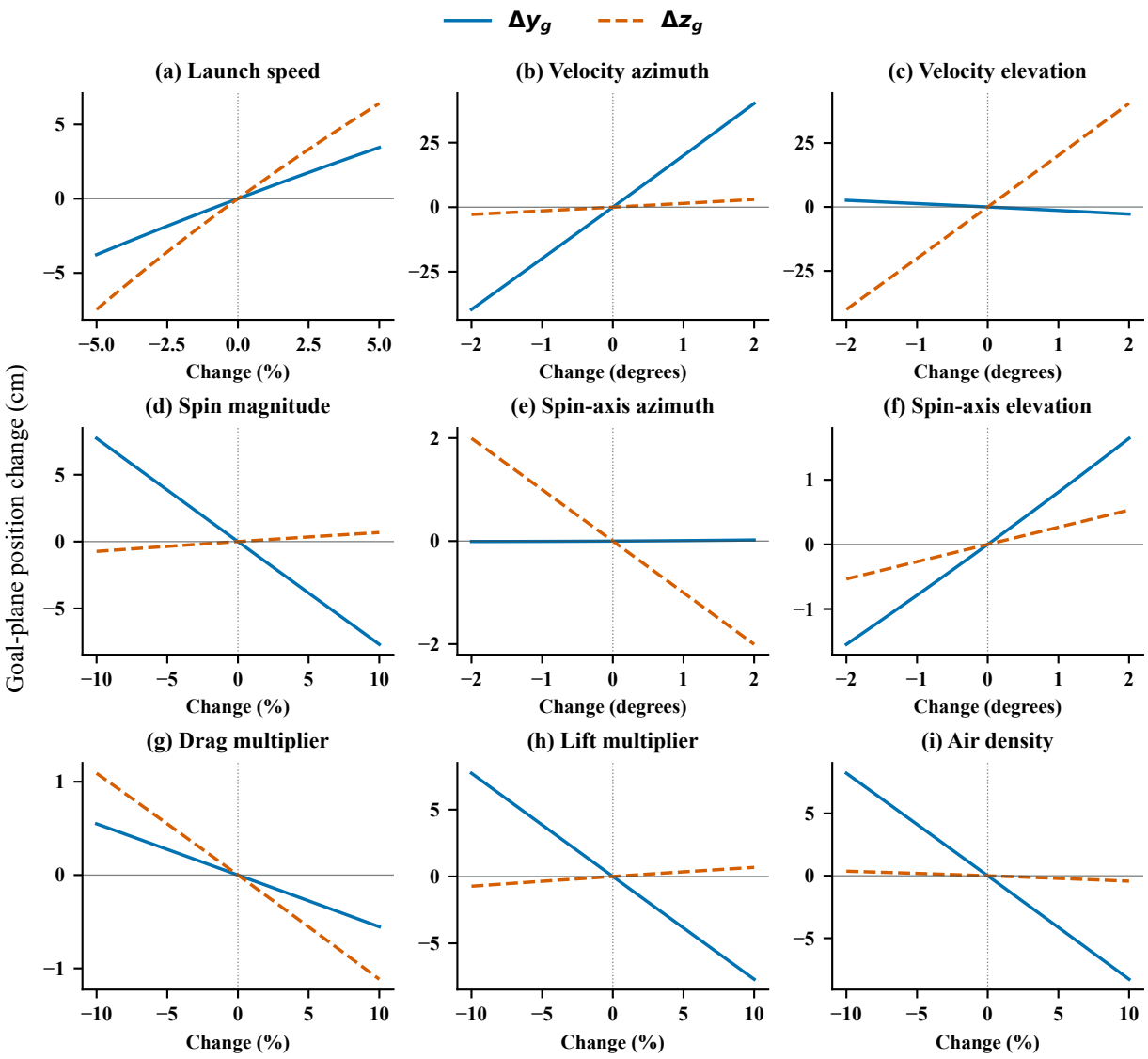


Figure 11: Goal-plane coordinate responses to the nine individual parameters.

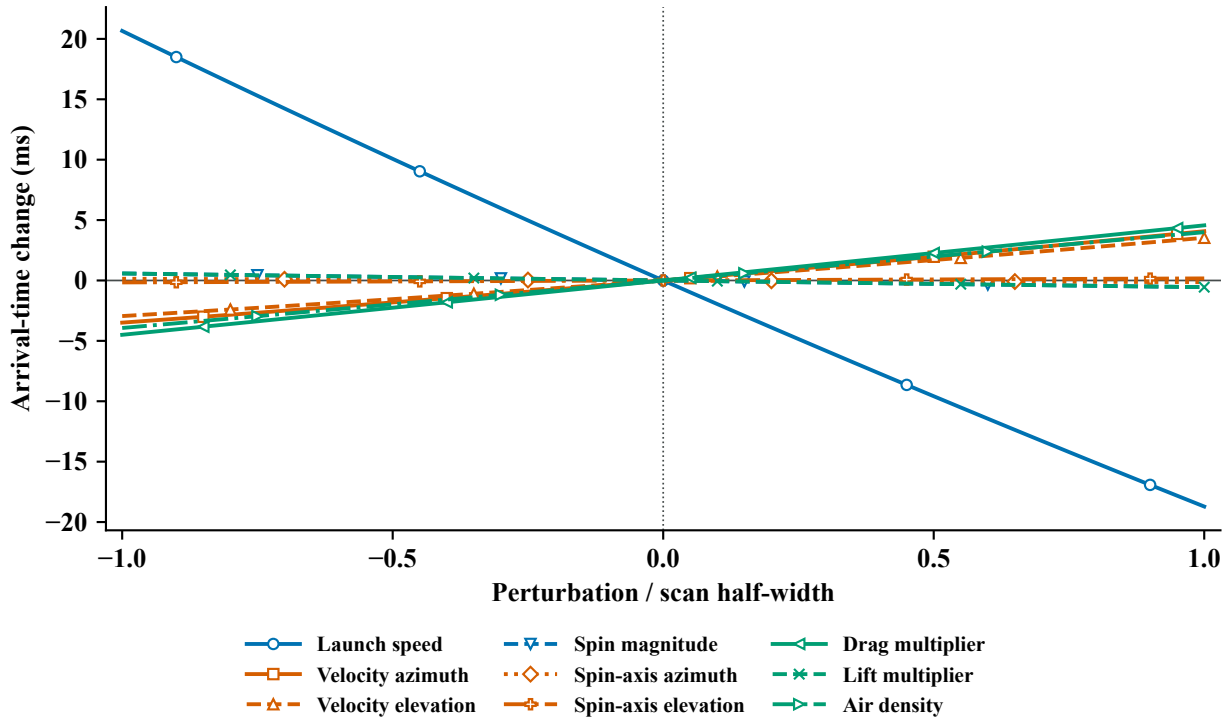


Figure 12: Arrival-time response, with perturbations normalized by each scan half-width.

Spin magnitude and α enter Magnus force as a product, so changes in opposite directions can compensate.

5.2 Crossing variation under early-position constraints

We scan a 17×17 grid of the two coefficient multipliers over $[1.2, 1.8]^2$, refitting velocity and spin components at each node. Total speed is at most 35 m/s, individual spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 0.002 m. The inner range is $[1.35, 1.65]^2$. Reported ranges describe the sampled feasible candidates.

Table 5: Ranges of goal-plane crossings compatible with the early positions.

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9122–2.0307	0.3973–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.9075–2.0368	0.3973–0.4043

The largest crossing displacement in the wider scan is 0.1052 m.

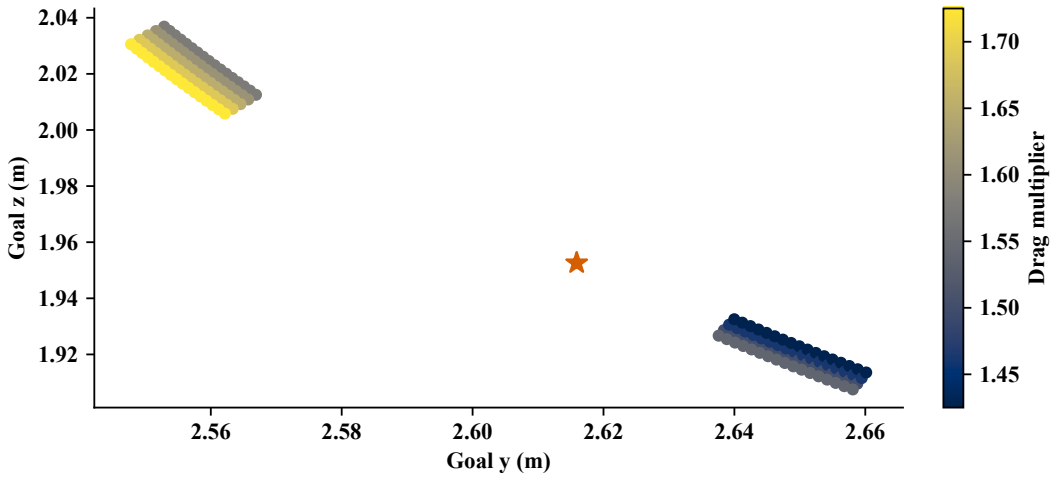


Figure 13: Scanned crossings with early residual at most 2 mm. Colour indicates the drag multiplier.

5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the individual scan ranges, with mass, radius and initial position fixed. Seed 20261108 generates 100000 trajectories; the early positions are allowed to change.

Table 6: Monte Carlo crossing positions and arrival times (100,000 trajectories).

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The crossing mean is $(2.6170, 1.9511)$ m and standard deviations are 0.2455 m and 0.2360 m. Under the ball-centre clearance rule, 95772 trajectories (95.77%) pass fully through the opening and 4228 pass above it.

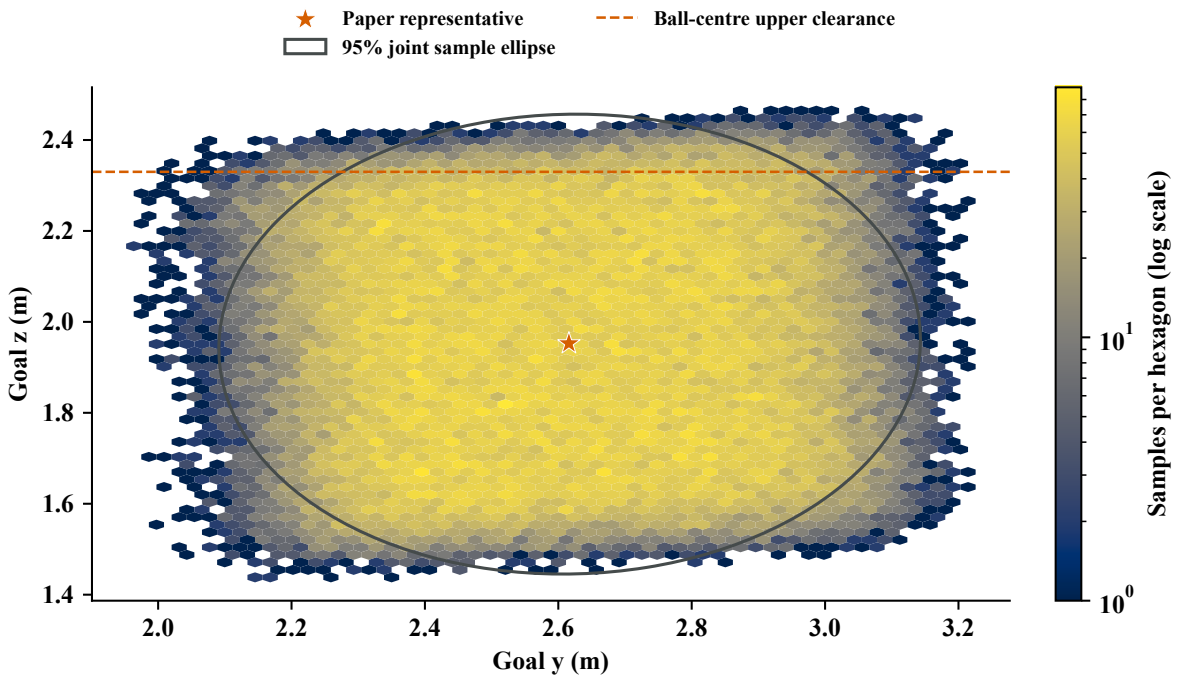


Figure 14: Joint-perturbation crossings; the ellipse covers 95% of simulated samples.

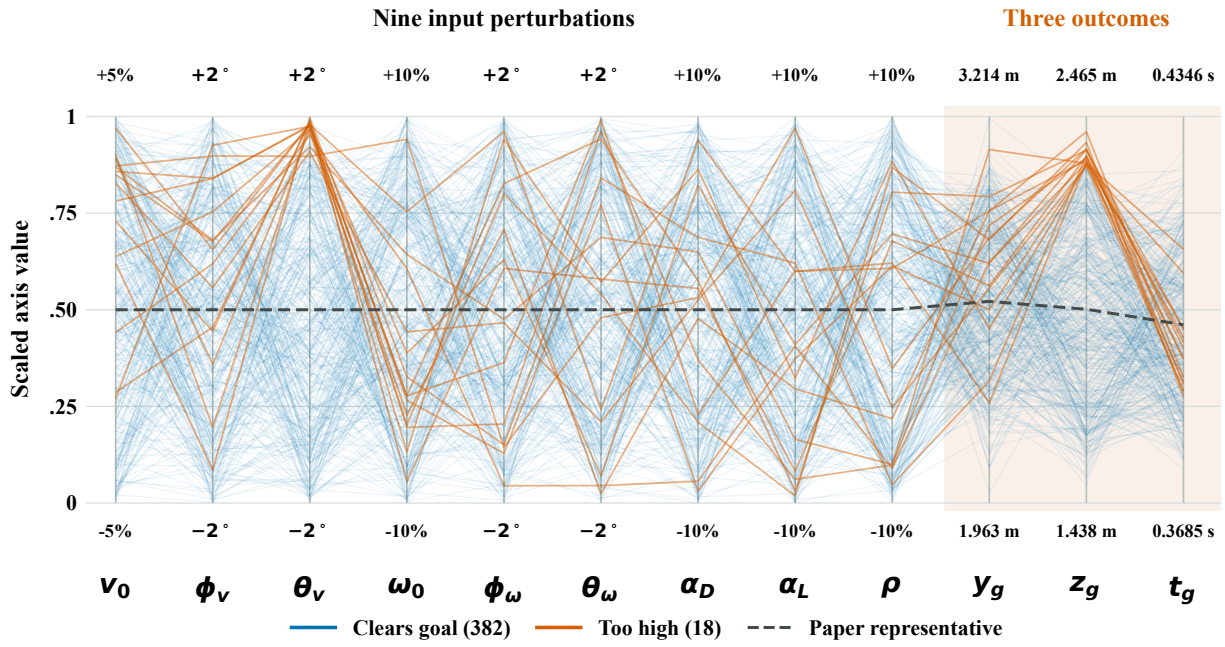


Figure 15: A random display of 400 flights connects the nine inputs to three outputs.

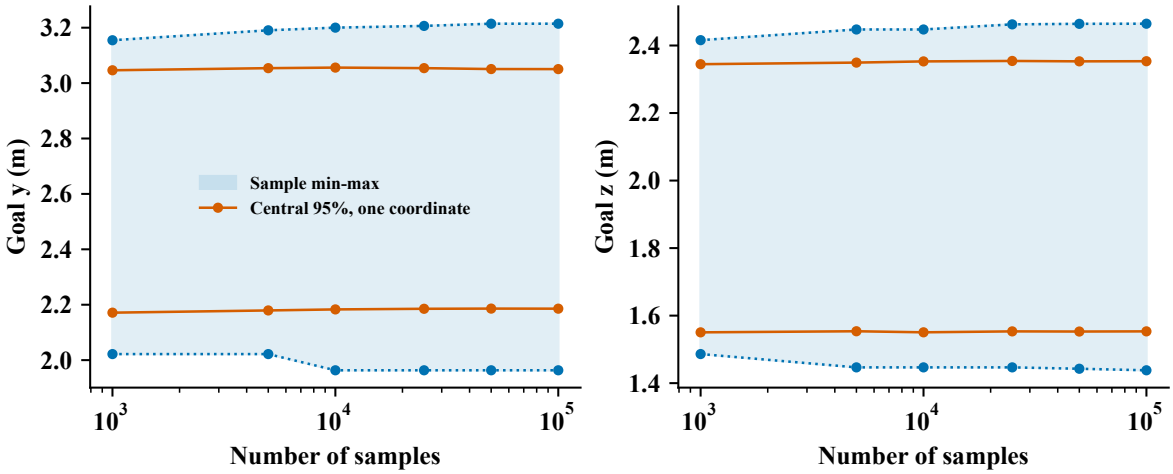


Figure 16: Effect of sample count on the simulated crossing distribution.

Fourth-order Runge–Kutta integration is compared with adaptive integration and a halved step. These ranges and fractions correspond to the stated uniform sampling assumptions.

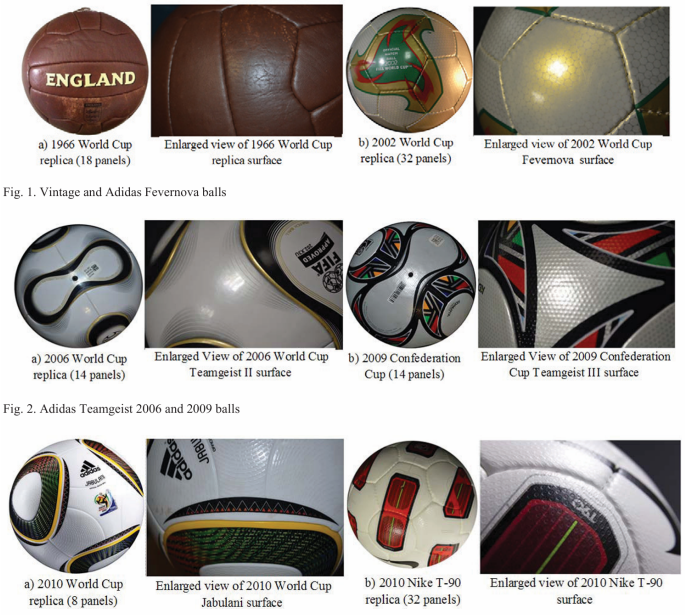
6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

A full-flight simulation compares realistic football geometry with a smooth sphere [9]. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.





(a) Illustrative seamed-football and smooth-sphere comparison [9].

(b) Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [16].

Figure 17: Football geometry, panels and surface textures.

Wind-tunnel experiments measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere [16]. Figure 17b shows the panels and surface textures of these footballs, while Figure 18 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.

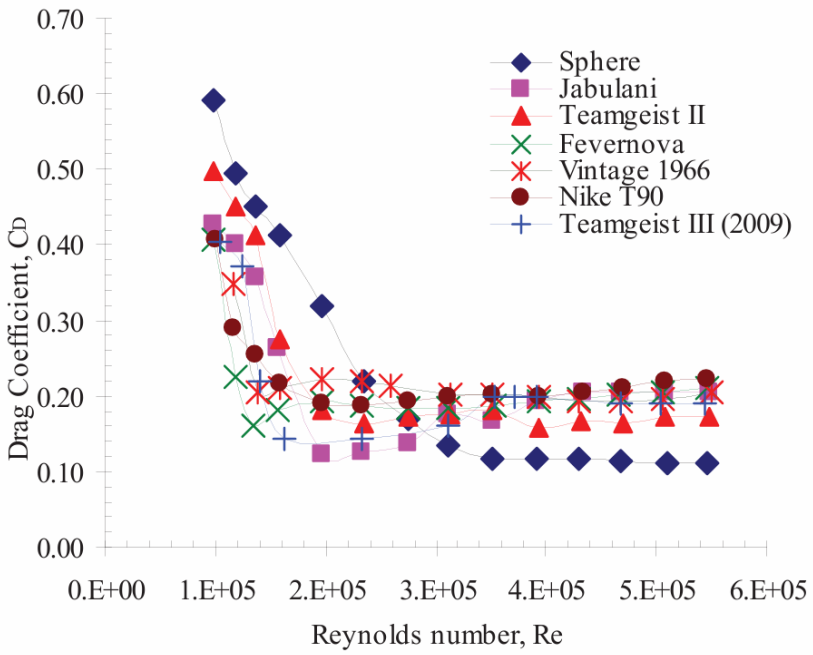


Figure 18: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of the source[16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. We use the high-Reynolds-number rotational-damping model described by Grad et al. and MuJoCo [13, 14], in which stronger spin produces stronger resistance. The angular-drag parameter is the software reference value $C_{\text{angular}} = 1.5$, with the thin-shell inertia approximation [8]. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

To assess its practical effect during a short flight, we hold the curved shot's initial velocity, spin and aerodynamic coefficients fixed, and compare constant spin with rotational damping. Figure 19 shows both the spin evolution and the resulting trajectory difference.

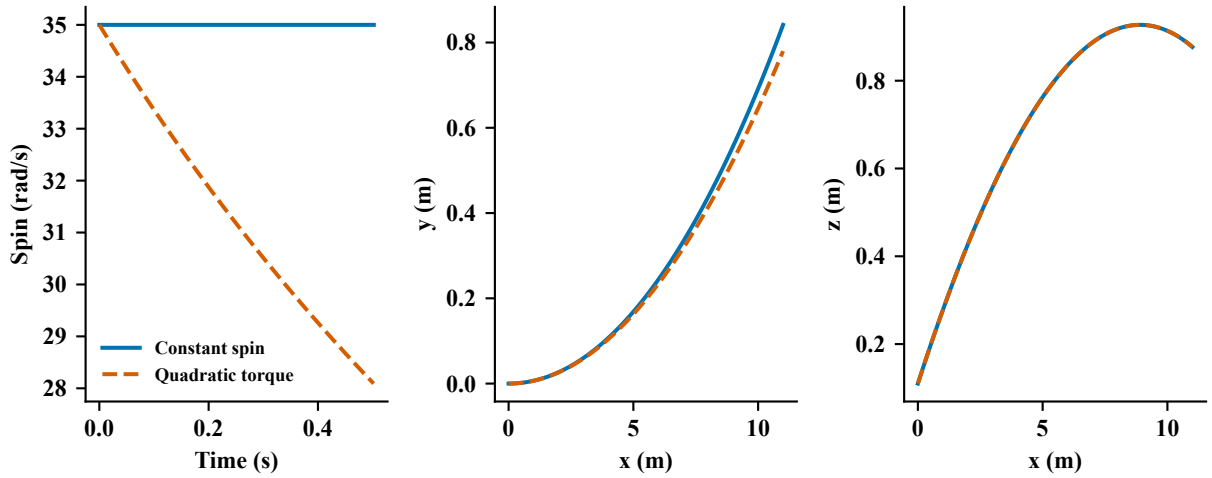


Figure 19: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [8]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{9}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 7 evaluates these rough estimates at the curved-shot launch speed. The listed wind speeds are illustrative conditions for comparison.

Table 7: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model separates the mechanisms of gravity, drag and Magnus lift. Gravity produces downward curvature, drag reduces speed, and spin redirects the path. We use constant drag and linear-spin lift.

Two cases are placement shots and the third is a power shot. Their M4 speeds at the selected start are 29.20, 28.34 and 32.05 m/s, respectively. Drag decreases as the ball slows, while spin direction determines the direction of Magnus force. The four-model comparison shows that fitting the early positions and predicting the goal crossing are distinct tasks: M4 has crossing errors of 0.113 m for the power shot and 0.146 m for the second placement shot.

Three actual penalties supply distinct early sampling sequences and held-out trajectories. Coordinate conversion places all three in the competition geometry. The 12 inverse outcomes connect the recorded motion to the force assumptions through spatial trajectories and the detailed power-shot comparison. For the competition-position example in Section 5, a one-degree launch-direction change moves the crossing by the displacement shown in the one-factor table. With the early positions fixed, the explored compatible states shift the crossing by up to 0.1052 m; joint perturbations are quantified by 100,000 Monte Carlo trajectories with fixed ball mass and radius.

8 Model Evaluation

8.1 Model Strengths

Physical interpretation. The model connects the path to three familiar effects: gravity bends it downward, drag reduces speed, and Magnus force redirects a spinning ball. The four-model comparison separates these mechanisms from a constant-acceleration description, making the differences between low placement shots and a higher power shot easier to explain.

A common framework for inference and prediction. Three early positions provide the input for estimating initial motion and continuing the flight with the same governing equations. Speed, force histories and goal-plane crossings can therefore be interpreted together. The common coordinate system and termination rules also make comparisons between models and recorded shots consistent.

Comparison with measurements. Published aerodynamic measurements support the coefficient choices, while later positions from three actual penalties assess the predicted continuation. The error table, trajectory plots and sensitivity study link the physical assumptions to observable differences. The wind and rotational-damping comparisons further identify when the simpler flight assumptions give useful approximations.

8.2 Model Weaknesses

Sensitivity of the inferred state. The fitted velocity, spin direction and aerodynamic parameters depend on the early tracking positions, frame selection and search bounds. Position differences over short time intervals strongly influence estimated acceleration, especially for flights close to the ground. The reference spin magnitude and the coupling between spin and lift parameter also affect the selected M4 state.

Simplified aerodynamics and environment. Constant coefficients during a flight smooth over flow transitions, panel orientation and fluctuating wakes. The spherical, still-air and constant-spin assumptions likewise average effects that may become more important for stronger winds, sustained spin or longer flight. Their relative importance changes with the ball and operating conditions.

Limited coverage of recorded conditions. The three penalties cover a modest range of shot types, balls and environments. Broader comparison would benefit from ball-specific aerodynamic data, directly measured spin and more early tracking positions. These additions would strengthen the initial-state estimate and help determine when wind and rotational damping should enter the main prediction.

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Appendix A: Data Sources, Frame Selection and Parameter Bounds

The three penalties come from the Sportec Open DFL dataset, which associates match events with TRACAB Gen5 multi-camera tracking at 25 Hz[6, 7]. The supplied CSV retains 39 source frames, with 15, 12 and 12 frames for the three events. Published positions are used without adding an observation-noise model.

Table 8: Correspondence between case labels and source records

Case	Event ID	Player	Selected samples
Placement 1	J03WMX	Dejan Ljubicic	3, 4, 5
Placement 2	J03WR9	Philipp Klement	3, 4, 5
Power shot	J03WOH	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. The height field is interpreted in metres as ball-centre height, and coordinates are converted with $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

The inversion bounds and force-correction settings are listed below.

Table 9: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\pi/2, \pi/2]$ rad
$C_D/C_D^*, \alpha/\alpha^*$	$[0.5, 1.5]$

Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
    return -factor*w


def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
```

```
--
    return record


def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
```